

A four-face die cannot belong to five permutation-fair dice

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Status. Complete elementary proof, independently reviewed. The symbolic determinant identity is checked by the accompanying exact script. Numerical exploration suggested the proof but is not used in it.

Theorem 1. *No permutation-fair collection of five dice contains a die with exactly four faces. Consequently, in conjunction with the universal $n - 1$ bound, every die in such a collection has at least five faces.*

We prove a stronger local statement using comparison moments only through degree three. No fourth-order comparison moment or global order condition is needed.

Lemma 2. *There do not exist four nondecreasing rational-valued functions $p_j : \{1, 2, 3, 4\} \rightarrow [0, 1]$, $j \in [4]$, such that*

$$\frac{1}{4} \sum_{q=1}^4 \prod_{j \in S} p_j(q) = \frac{1}{|S| + 1} \quad (S \subseteq [4], |S| \leq 3).$$

Proof. Put $y_j = 2p_j - 1$. The assumed moments imply

$$\mathbb{E}y_j = 0, \quad \mathbb{E}(y_i y_j) = \frac{1}{3} \quad (i \neq j), \quad \mathbb{E}(y_i y_j y_k) = 0 \quad (i, j, k \text{ distinct}),$$

where expectation is uniform on the four ordered values of q . Use the orthonormal basis of the mean-zero subspace consisting of the columns of

$$H = \begin{pmatrix} -1 & -1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix}, \quad \frac{1}{4} H^T H = I_3.$$

Write $y_j = H(A_j, B_j, C_j)^T$. All coefficients are rational. The three adjacent differences of y_j are

$$2(B_j + C_j), \quad 2(A_j - B_j), \quad 2(B_j - C_j).$$

Monotonicity is therefore equivalent to

$$A_j \geq B_j \geq |C_j|. \tag{1}$$

Moreover $A_j > 0$, since $A_j = 0$ would make $y_j = 0$, contradicting its nonzero pairwise inner products. Orthonormality gives

$$A_i A_j + B_i B_j + C_i C_j = \frac{1}{3} \quad (i \neq j). \tag{2}$$

Put $b_j = B_j/A_j \geq 0$ and $c_j = C_j/A_j$. The product of the three distinct columns of H is identically -1 ; all other third-order products of its columns have expectation zero. Consequently the triple-moment condition, divided by the positive product $A_i A_j A_k$, is

$$c_i(b_j + b_k) + c_j(b_i + b_k) + c_k(b_i + b_j) = 0 \quad (i, j, k \text{ distinct}).$$

Collect the four equations according to their omitted index. They say $M(b)c = 0$, where

$$M(b)_{ii} = 0, \quad M(b)_{ij} = \sum_{k \notin \{i,j\}} b_k \quad (i \neq j).$$

Writing e_r for elementary symmetric functions, direct expansion gives

$$\det M(b) = -4(e_1(b)e_3(b) - 4e_4(b)) = -4 \sum_{\substack{i,j,k \text{ distinct} \\ j < k}} b_i^2 b_j b_k. \quad (3)$$

The sum on the right has twelve terms. If at least three of the b_j are positive, it is strictly positive, so $M(b)$ is invertible. Hence all $C_j = 0$.

In this case the four rational vectors $v_j = (A_j, B_j) \in \mathbb{Q}^2$ have pairwise inner products $1/3$. Their Gram matrix is

$$G = D + \frac{1}{3}\mathbf{1}\mathbf{1}^T, \quad D_{jj} = A_j^2 + B_j^2 - \frac{1}{3}.$$

Since $\text{rank } G \leq 2$, the diagonal matrix D has rank at most three. Thus some $A_j^2 + B_j^2 = 1/3$. This is impossible over $\mathbb{Q}$: clearing denominators in $a^2 + b^2 = 1/3$ gives $3(u^2 + v^2) = w^2$ for coprime integers u, v, w . It first forces $3 \mid w$, then $u^2 + v^2 \equiv 0 \pmod{3}$, so $3 \mid u, v$, a contradiction.

It remains to consider at most two positive b_j . At least two indices, say i and j , have $B_i = B_j = 0$, and (??) also gives $C_i = C_j = 0$. Choose a third index k . Equation (??) gives

$$A_i A_j = A_i A_k = A_j A_k = \frac{1}{3}.$$

Positivity implies $A_i = A_j = A_k$ and $A_i^2 = 1/3$, again impossible for rational A_i . Both cases contradict the assumed functions. $\square$

Proof of Theorem ??. Suppose that the distinguished die D_0 has four faces, in increasing order. For the other four dice put $p_j(q) = \Pr(D_j < D_0(q))$. These functions are rational and nondecreasing. Permutation fairness, restricted to D_0 and any set S of the other dice, gives

$$\frac{1}{4} \sum_q \prod_{j \in S} p_j(q) = \Pr(D_j < D_0 \text{ for every } j \in S) = \frac{1}{|S| + 1}.$$

Lemma ?? is a contradiction. $\square$

Consequence for the exceptional-die optimum. For $g(n)$, the minimum number of faces of a prescribed die with all other dice unrestricted, the independently established results now read

$$g(2) = 1, \quad g(3) = 2, \quad g(4) = 3, \quad g(5) \geq 5, \quad n - 1 \leq g(n) \leq Cn^2.$$

In particular, the tempting equality $g(n) = n - 1$ for every n is false. The exact value of $g(5)$ and the asymptotic order remain open here.